

ENDORESECTION TECHNIQUE WITH/WITHOUT BRACHYTHERAPY FOR MANAGEMENT OF HIGH POSTERIOR CHOROIDAL MELANOMA

Extended Follow-up Results

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Purpose: To evaluate the long-term efficacy and safety of endoresection for high posterior choroidal melanoma.

Methods: Retrospective nonrandomized interventional case series. Forty-one patients had endoresection as primary treatment for posterior choroidal melanoma. Of these, 21 patients had adjuvant brachytherapy. The inclusion criteria were tumor thickness ≥ 8 mm, base diameter < 15 mm, and posterior tumors not extending anterior to the equator. Main outcomes measures were enucleation rate, visual outcome, surgical complications, local recurrence, metastasis, and mortality.

Results: Mean follow-up time was 102.5 months. Mean preoperative best-corrected visual acuity was 20/100. Mean tumor thickness was 9.8 mm (range, 7.7–13.5 mm; standard deviation, 1.7 mm), mean base diameter was 9.9 mm (range, 5–15 mm; standard deviation, 2.8 mm). At the latest visit, 36 patients (87.8%) still retained the eye. Mean postoperative best-corrected visual acuity was 20/1,625. Retinal detachment was the main postoperative complication (28.9%). At completion of follow-up, 12% of patients had phthisis bulbi, and 3 developed chronic hypotony. Five patients (12.2%) had local tumor recurrence; none of them had received brachytherapy as initial treatment. At 5 years of follow-up, 3 patients (7.3%) had liver metastasis. On Kaplan–Meier analysis at 10 years, all-cause mortality was 7.3% and specific mortality because of melanoma was 2.4%.

Conclusion: Endoresection of high posterior melanomas was not associated with a higher risk of metastasis, death, or local recurrence than other reported techniques used to treat similar melanomas.

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Management options for posterior uveal melanoma have evolved from enucleation as the sole valid possibility to an armamentarium of options, including enucleation, laser photocoagulation, transpupillary

thermotherapy, irradiation with plaques and charged particles, and local resection.^{1–6}

Internal resection of uveal melanoma by ab interno retinochoroidectomy was first reported by Peyman and Cohen⁷ for tumors within 1 or 2 disk diameters of the optic nerve head. Later on, Lee et al⁸ and Damato et al⁹ reported results of the same technique with a longer follow-up and a larger patient cohort. Damato was the first to coin the term *endoresection*.

The validity of endoresection in the management of posterior uveal melanomas is highly debated because of concerns that surgical manipulation of the tumor might result in systemic and local dissemination of

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malignant cells. Ocular oncologists might consider performing endoresection as a primary intervention or as a salvage procedure whenever favorable response to other conventional primary or salvage procedures is deemed unlikely.^{9,10} Indications include rapidly growing posterior melanomas whose thickness is greater than their base diameter as an alternative to enucleation to evade morbidity associated with radiotherapy,¹¹ local tumor recurrence after brachytherapy, transpupillary thermotherapy, or photocoagulation, and to conserve vision in patients with a single eye.^{9,10}

In this report, we present the extended follow-up results of primary and salvage endoresection for high posterior choroidal melanoma. The surgical technique and the safety and efficacy of endoresection are discussed.

Materials and Methods

The study includes all patients with high posterior choroidal melanoma who underwent surgery in our department between 1997 and 2012. Four patients operated from 2012 to the time of writing were excluded because of the short follow-up time. All these more recent patients are alive, had no surgical complications, and show no signs of metastasis. All procedures were performed by the same surgeon (J.G.-A.). Inclusion criteria were tumor thickness ≥ 8 mm, base diameter < 15 mm, and posterior tumors not extending anterior to the equator. The only patient whose tumor thickness was < 8 mm (7.7 mm) had a colon neoplasm with hepatic metastasis at the time of the diagnosis and preferred endoresection. All patients received a thorough explanation of the controversial nature of the procedure and signed an informed consent before enrollment in the study.

Preoperative work-up included Snellen best-corrected visual acuity, complete ophthalmologic examination, A-scan and B-scan with measurement of maximum thickness and base diameter of the tumor, fundus photography, and fluorescein angiography. Orbit imaging was done in cases where orbital dissemination was suspected. Screening for metastasis was done in all cases, and included serum biochemistry and liver function tests, ultrasound study of the liver, chest radiography, and bone scanning.

The surgical technique varied according to the extent of retinal involvement. Because the tumor had not yet invaded the retina, we performed 20-g vitrectomy with posterior hyaloid dissection, 120° anterior retinotomy, and endolaser photocoagulation using an 810-nm diode laser at 800 to 1,000 mW. Laser treatment was extended to 2 mm beyond the tumor margins. The melanoma was

then removed with the vitrectomy probe using a bimanual technique. Intraocular pressure (IOP) was increased to 100 mmHg for 5 minutes to prevent bleeding from choroidal vessels. Systemic hypotension was not induced during the procedure. Excision began at the apex of the tumor and carried down to the scleral bed. Diode endolaser (800–1,000 mW) was applied to the scleral bed to destroy residual tumor cells. The retina was reattached with perfluorocarbon liquid and air. Retinopexy was performed by applying endolaser photocoagulation to the retinotomy edges followed by fluid–air exchange and silicone oil–air exchange. In case the tumor had already invaded the retina, diode laser was applied throughout the retina, and the tumor and the retina were removed together. Silicone oil was removed after 3 months.

Since January 2003, we have treated all cases with adjunctive brachytherapy after surgery to prevent local recurrence, except in one patient who refused brachytherapy. A ruthenium 106 plaque (¹⁰⁶Ru) was sutured to the sclera to cover as much as possible of the surgical coloboma. The plaque was removed after delivering 80 Gy to a depth of 3 mm.

All patients were instructed to maintain a face-down position for 1 week. Follow-up examination was performed at 1 week, 1 month, 3 months, and every 6 months thereafter. Ophthalmoscopic examination and screening for metastasis were carried out every 6 months.

Retrospectively, in available cases, fluorescence in situ hybridization was carried out on paraffin section of tumor endoresection samples. We used SOCEP 3 Spectrum orange Vysis (32-130003) and SGCEP8 spectrum green Vysis (32-132008). In all cases, 400 cells were counted, and the cutoff limit for deletion and amplification was 20%.

Patient data were analyzed with SSPS 12.0 for Windows. Kaplan–Meier survival curves for 10 years were calculated and statistically compared by log-rank analysis.

Results

Patient Characteristics

The study included 41 patients, 27 men and 14 women. Mean age was 53.6 years (range, 26–82 years; standard deviation, 12.4 years). Mean follow-up time was 102.5 months (range, 20–180 months; standard deviation, 44.8 months). Concurrent disease included diabetes mellitus (1 case), hypertension (3 cases), schizophrenia (1 case), colon carcinoma with hepatic metastasis (1 case), treated breast carcinoma (1 case), and treated prostate carcinoma (1 case) (Table 1).

Table 1. Baseline Patient Characteristics

Baseline Characteristics	N (%)
Male	27 (66)
Female	14 (34.1)
Age, years	
<40	5 (12.2)
40–50	10 (24.3)
51–60	14 (34.1)
>60	12 (29.2)
Tumor quadrant	
Juxtapapillary	5 (12.2)
Temporal	16 (39)
Nasal	4 (9.8)
Superior	2 (4.9)
Superotemporal	6 (14.6)
Superonasal	1 (2.4)
Inferior	2 (4.9)
Inferotemporal	2 (4.9)
Inferonasal	3 (7.3)
Maximum tumor basal diameter, mm	
<8	12 (29.2)
8–10	10 (24.4)
10.1–12	8 (19.5)
>12	11 (26.8)
Tumor height, mm	
<8	1 (2.4)
8–10	24 (58.5)
10.1–12	14 (34.1)
>12	2 (4.9)
Visual acuity	
20/20–20/25	3 (7.3)
<20/25–20/40	14 (34.1)
<20/40–20/200	16 (39)
<20/200	8 (19.5)
Follow-up, months	
<24	1 (2.4)
24–48	4 (9.75)
>48–60	5 (12.2)
>60	31 (76)

Endoresection was performed as primary treatment in all cases. Twenty patients (48.8%) had endoresection alone, whereas 21 patients (51.2%) had endoresection and adjuvant brachytherapy. Mean preoperative best-corrected visual acuity was 20/100 (range, hand movement–20/20). Mean tumor thickness was 9.8 mm (range, 7.7–13.5 mm; standard deviation, 1.7 mm), whereas mean base diameter was 9.9 mm (range, 5–15 mm; standard deviation, 2.8 mm).

The tumor was in a juxtapapillary location in 5 patients (12.2%), temporal in 16 patients (39%) (Figure 1), nasal in 4 patients (9.8%) (Figure 2), superior in 2 patients (4.9%), superotemporal in 6 patients (14.6%) (Figure 3), superonasal in 1 patient (2.4%), inferior in 2 patients (4.9%), inferotemporal in 2 patients (4.9%), and inferonasal in 3 patients (7.3%) (Figure 1, A and B; Figure 2, A–D; Figure 3, A and B).



Fig. 1. **A.** Composite color fundus photograph of the right eye of a 44-year-old patient. A round, elevated yellowish-brown choroidal melanoma is visible in the temporal fundus. Intraretinal and subretinal hemorrhage can be seen on the tumor surface. The tumor has a largest base diameter of 6.24 mm and an apical height of 8.05 mm. Preoperative best-corrected visual acuity was 20/80. **B.** Fundus appearance 1 year postoperatively after primary endoresection and brachytherapy shows surgical coloboma and retinal pigment epithelium changes along the margins. Postoperative best-corrected visual acuity remained the same (20/80).

Thirty-six patients presented with exudative retinal detachment, which was bullous in 5 of them. A choroidal mass was the presenting sign in the remaining 5 patients. Four melanomas were amelanotic. Signs of malignancy were unequivocal in all patients. Corroborating immunohistochemistry evidence of malignancy was available in 39 of the 41 patients. Seven patients were lost to follow-up after an average of 80.3 months.

Anatomical and Functional Outcomes

At the latest visit, 36 of 41 patients (87.8%) still retained the eye, and 5 patients had undergone

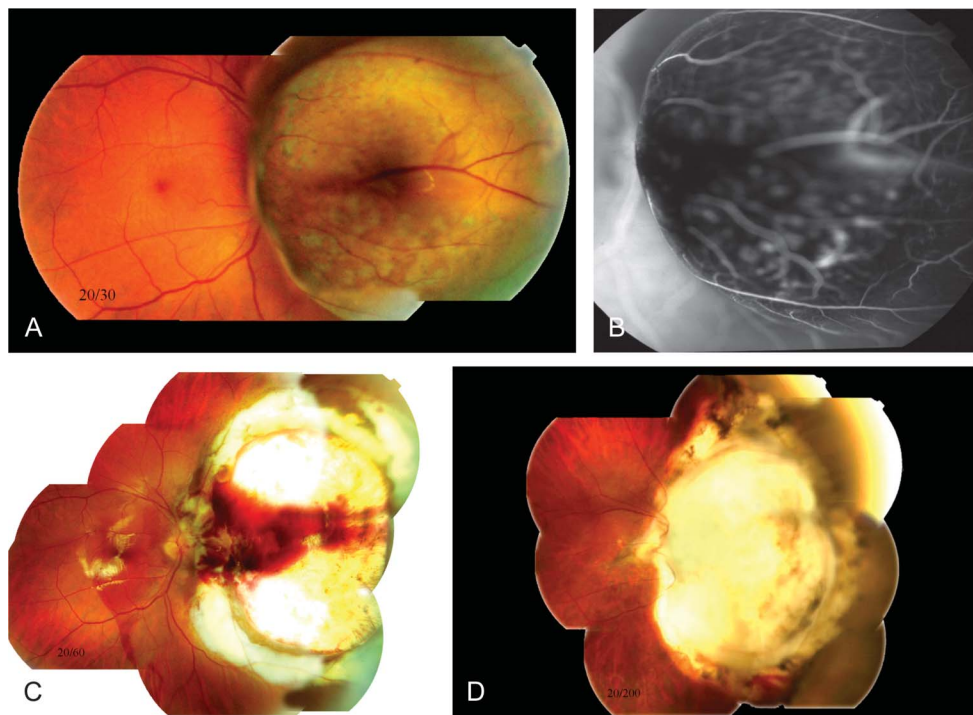


Fig. 2. **A.** Composite color fundus photograph of the right eye of a 39-year-old patient. A markedly elevated, round choroidal melanoma is occupying almost the entire nasal fundus and has obscured visualization of the optic disk. The tumor surface shows irregular grayish pigmentation. The tumor has a largest base diameter of 12.7 mm and an apical height of 11.8 mm. Preoperative best-corrected visual acuity was 20/30. **B.** Venous phase fluorescein angiography shows the independent tumor vascular network with some dye leakage. The retinal vessels in the background appear blurred because of tumor elevation. **C.** Fundus appearance 1 month after primary endoresection without brachytherapy. Blood has accumulated in the scleral bed of the resected tumor and has trickled around the optic disk and into the macular area forming submacular hemorrhage. Postoperative best-corrected visual acuity was 20/60. **D.** Fundus appearance 1 year after surgery shows the surgical coloboma and retinal pigment epithelium changes along the margins. Consecutive optic atrophy and macular scarring have developed. Best-corrected visual acuity was 20/200.

enucleation. Indications for enucleation were local recurrence in 4 patients and phthisis bulbi in 1 patient. Final best-corrected visual acuity ranged from no light perception to 20/20, with mean 20/1,625 as follows: $\leq 20/400$ in 21 patients, 20/200 to 20/100 in 12 patients, and $>20/100$ in 8 patients.

Surgical Complications

The main postoperative complication was retinal detachment after silicone oil removal. Among the 38 patients who underwent oil removal, 11 (28.9%) had retinal detachment. Postoperative retinal detachment resulted from a break at one of the margins of the surgical coloboma in all patients except one who had a tear in peripheral lattice degeneration. Five patients developed proliferative vitreoretinopathy that was localized to one quadrant. Those patients were managed by scleral buckling combined with pars plana vitrectomy and long-acting perfluoropropane (C_3F_8). None of them developed recurrent detachment until the end of follow-up period. The remaining 3 patients developed postoperative hypotony without retinal detachment. Silicone oil was not removed in these 3 cases.

Other postoperative complications included bleeding from the scleral bed occurring in all cases (100%), early postoperative ocular hypertension in 14 patients

(34.1%), epiretinal membrane formation in 5 patients (12.2%), postradiation retinopathy in 4 patients (9.8%), subretinal fibrosis in 1 patient (2.4%), and cystoid macular edema in 1 patient (2.4%). Two patients (4.9%) developed subretinal neovascularization 2 years after endoresection. Thirty-nine patients needed cataract surgery during the follow-up period, either concomitantly with silicone oil removal performed 3 to 6 months after endoresection or as a third procedure later on. Five patients (12.2%) ended up with phthisis bulbi (Table 2).

Local Recurrence, Metastasis, and Mortality

Local tumor recurrence occurred in 5 patients (12.2%) in the group who had endoresection alone as the initial treatment. Two of these patients (4.9%) developed recurrence before completing 3 years of follow-up. Four of the five had enucleation, whereas the remaining one was treated by brachytherapy. None of the patients whose initial regimen consisted of endoresection and adjuvant brachytherapy had local recurrence. Conjunctival or orbital recurrences were not detected in any case.

At the start of the study, 1 patient presented with liver metastasis from colon carcinoma (Case 30). No new metastasis was detected at 3 years of follow-up. At 5 years of follow-up, a total of 3 patients (7.3%)

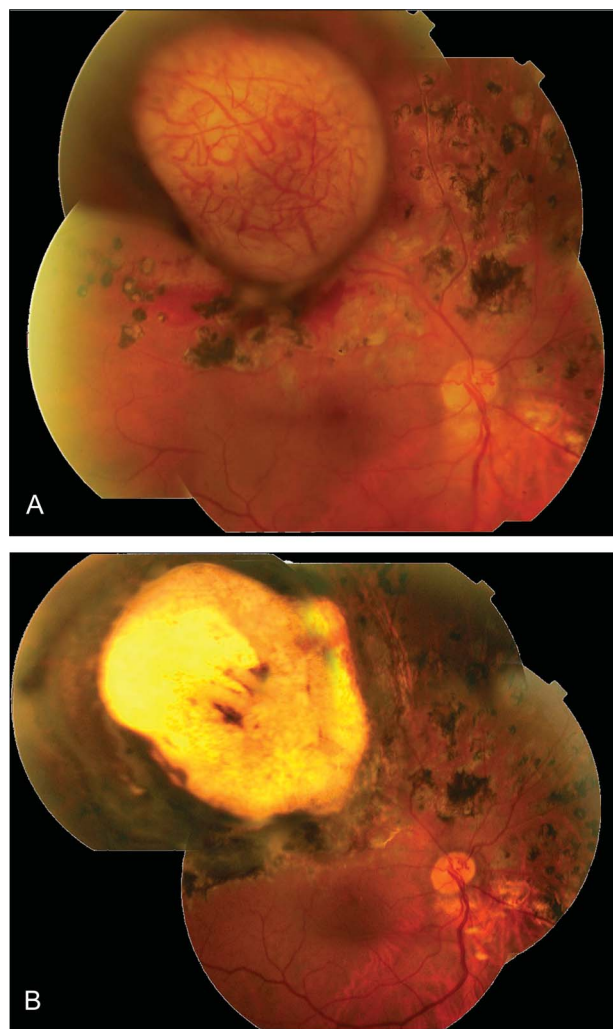


Fig. 3. **A.** Composite color fundus photograph of the right eye of a 59-year-old patient. A dome-shaped, markedly elevated amelanotic choroidal melanoma has developed in the superotemporal quadrant at the equator. The tumor's posterior margin extends into the posterior pole. The tumor's largest base diameter was 11 mm and apical height was 8 mm. The choroidal vascular network is clearly visible within the tumor. Preoperative best-corrected visual acuity was 20/40. **B.** Fundus appearance 1 year after primary endoresection and brachytherapy. Confluent old laser marks and retinal pigment epithelium changes extend well beyond the margins of the surgical coloboma. Postoperative best-corrected visual acuity was 20/200. The patient had an old episode of hemicentral retinal vein occlusion that was treated by scatter laser photocoagulation.

had liver metastasis including (Case 30). These additional 2 patients already had carcinoma elsewhere at the time they were diagnosed with melanoma (treated breast carcinoma and treated prostate carcinoma). One of them died at 63 months after endoresection. The other patient was lost to follow-up at 70 months after surgery. The patient who had hepatic metastasis from colon carcinoma (Case 30) died at 39 months after endoresection. Mean survival time for all patients was 170 months (95% confidence interval,

Table 2. Postoperative Complications

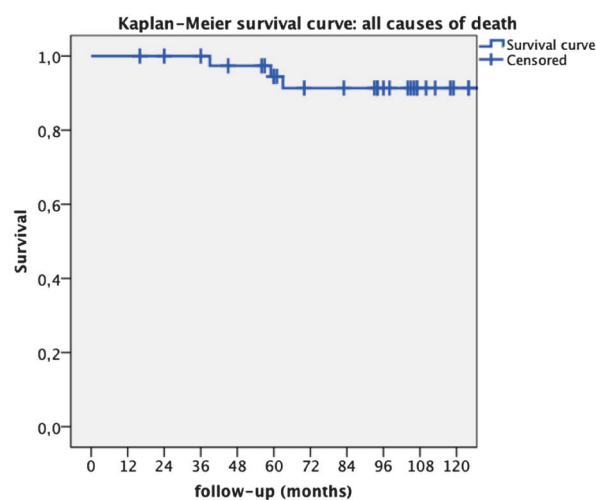
Postoperative Complications	N (%)
Retinal detachment	11 (28.9)
Phthisis bulbi	5 (12.2)
PRR	4 (9.8)
ERM	5 (12.2)
CNVM	2 (4.9)
Subretinal fibrosis	1 (2.4)
Ocular hypertension	14 (34.1)
VH	1 (2.4)
CME	1 (2.4)
Corneal decompensation	1 (2.4)

CME, cystoid macular edema; CNVM, choroidal neovascular membrane; ERM, epiretinal membrane; PRR, postradiation retinopathy; VH, vitreous hemorrhage.

159.3–180.8). The Kaplan–Meier survival analysis for all causes at 10 years was 92.7%. Specific survival for melanoma at 10 years was 97.6% (Figures 4 and 5, respectively). No significant differences in 10-year survival were found between patients who received adjuvant brachytherapy and those who did not ($P = 0.1$) (Figure 6). Thirty-three patients completed at least 5 years of follow-up. All-cause survival in this group was 94%, whereas melanoma-specific survival was 97%. Twenty-two patients completed 8.5 years of follow-up. In this group, all-cause survival and melanoma-specific survival were 100% (Table 3, Figures 4–8).

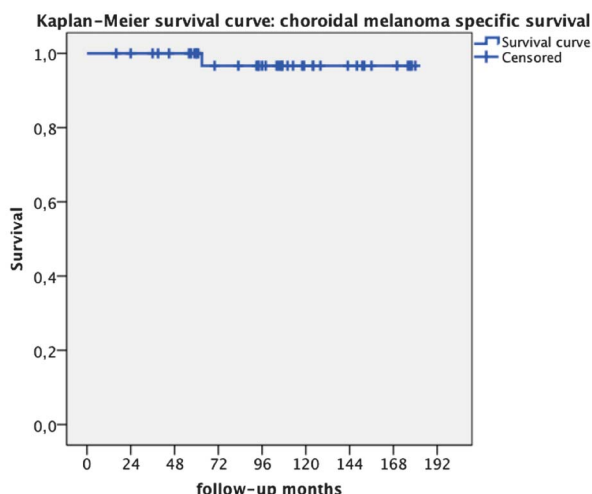
Chromosomal Analysis

Chromosomal analysis was available in 21 cases. Chromosome 3 loss was detected in 5 cases examined



Follow-up (months)	12	24	36	48	60	72	84	96	108	120
Patients at risk	41	40	39	36	33	28	27	24	17	13

Fig. 4. Kaplan–Meier visual survival curve: all causes of death.



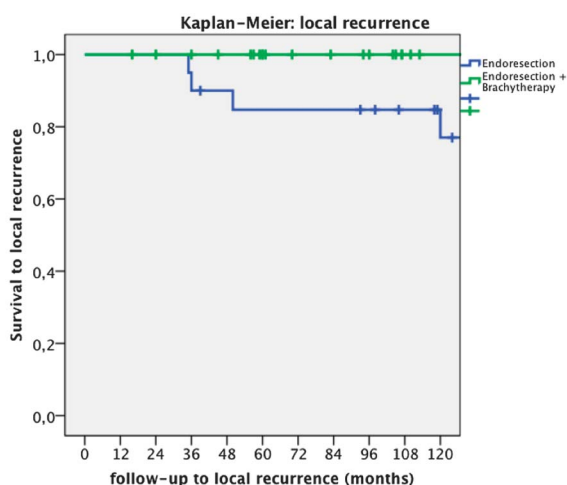
Follow-up (months)	12	24	36	48	60	72	84	96	108	120
Patients at risk	41	40	39	36	33	28	27	24	17	13

Fig. 5. Kaplan-Meier visual survival curve: choroidal melanoma-specific survival.

(23.8%). Polyploid Chromosome 8 was present in 3 cases. No patient had both abnormalities.

Discussion

In the endoresection technique used in this study, we adopted certain measures that we believe were decisive in achieving the relatively low rates of recurrence and



Follow-up (months)	12	24	36	48	60	72	84	96	108	120
Patients at risk (Endoresection + Brachytherapy)	20	18	17	16	13	9	8	7	2	0
Patients at risk (Endoresection)	21	21	20	18	17	17	17	16	14	11

Fig. 6. Kaplan-Meier visual survival curve: local recurrence.

metastasis as compared with other series. These included application of diode laser up to 2 mm beyond the tumor margins before resection to close the tumor's feeding vessels and decrease the risk of hematogenous dissemination and bleeding during tumor removal, raising the IOP to close the choroidal circulation, applying diode laser to the surgical coloboma bed, fluid-air exchange after tumor removal to eliminate residual cells in the vitreous cavity, lowering IOP during removal of instruments through sclerotomies, and conjunctival and scleral examination along with thorough washing at the conclusion of the procedure. Adjuvant cryotherapy was not applied to the sclerotomies. We preferred to perform endoresection using 20-gauge instruments in all cases because of their higher efficacy in removing the tumor than smaller-gauge instruments.

Damato et al⁹ reduced the ocular blood flow by lowering the systolic blood pressure to 50 mmHg or 60 mmHg. We believe that raising the IOP is the safest way to close the choroidal circulation, because it does not affect cerebral blood flow, and therefore avoids life-threatening complications. Experimental studies using animal models have shown that irreversible retinal damage occurs at 97 minutes after central retinal artery occlusion.¹² In our technique, IOP is elevated for approximately 5 minutes.

Bechrakis et al¹³ and Bornfeld et al¹⁴ found that tumor size correlates with mortality and morbidity, that is, larger tumor base diameter correlates with increased patient mortality, whereas greater tumor thickness correlates with increased ocular morbidity. Our series included 41 primary cases. Eighteen patients had large tumors and 23 had medium-sized tumors according to the Collaborative Ocular Melanoma Study (COMS) criteria. In these 2 subgroups, melanoma-specific mortality at 5 years was 0%.

In terms of tumor thickness, the present series had a mean tumor thickness of 9.8 mm; therefore, it can be considered as a large melanoma series. In the COMS large tumor trial, the estimated all-cause mortality was nearly 40% at 5 years while the estimated tumor-specific mortality was 27%.^{15,16} In comparison, the 10-year all-cause mortality in our series was 7.3% and melanoma-specific mortality was 2.4%. Subgroup analysis showed that 33 patients completed at least 5 years of follow-up. All-cause mortality in this group was 6%, whereas melanoma-specific mortality was 3%. Twenty-two patients completed 8.5 years of follow-up. In this group, all-cause mortality and melanoma-specific mortality were 0%.

The variable presentation of choroidal melanoma renders the comparison between our series and the COMS large tumor trial difficult. Patients enrolled in the COMS large tumor trial had a mean age of

Table 3. Local Recurrence, Metastasis, Enucleation, and Death

Local Recurrence, Metastasis, Enucleation, and Death	N (%)	Kaplan–Meier Outcome (Patients at Risk)				
		12 Months	36 Months	60 Months	96 Months	120 Months
Local recurrence	5 (12.2)					
Endoresection + brachytherapy	0 (0)	20	17	13	7	0
Endoresection alone	5 (12.2)	21	20	17	16	11
Metastasis	2 (4.9)	—	—	—	23	13
Enucleation	4 (9.8)	—	—	—	23	11
Death	1 (2.4)	41	39	33	24	13

60 years versus 53.6 years in our series. The mean tumor base diameter in COMS was greater than in our study (17.2 vs. 9.9 mm, respectively). Nevertheless, mean tumor thickness was comparable (9.5 vs. 9.8 mm).

In terms of tumor base diameter, our results are comparable with those of the COMS medium tumor trial, while bearing in mind that mean tumor thickness in that trial was 4.8 mm and mean base diameter 11.4 mm. The estimated 5-year all-cause mortality and disease-specific mortality for the COMS tumors were 20% and 10%, respectively. These data indicate that this study had higher survival rate, and lower metastasis and local recurrence rates in relation to tumor size as compared with the COMS large and medium tumor trials.

The risk of iatrogenic dissemination of tumor cells during endoresection because of incomplete tumor removal has always been the major concern regarding this technique. En bloc removal of malignant tumors is considered to have a higher safety profile.⁶ This notion, however, is challenged by the accumulated clinical evidence that endoresection is not associated with higher local recurrence and metastasis rates as compared with other modalities.

In this study, local recurrence was found to be 5.1% and 6% at 3 years and 5 years, respectively. Damato et al⁹ reported 2 cases of recurrent disease in 41 primary endoresections, and 1 after salvage endoresection. Kertes et al¹⁷ reported 1 local recurrence in 32 patients with a mean follow-up of 3.5 years. Karkhaneh et al¹⁸ detected 1 recurrence at the margins of the surgical coloboma and another new tumor focus in 20 patients with a mean follow-up time of 89.5 months.

Metastasis occurred in 3 of our patients at 5 years of follow-up, although one of those patients had liver metastasis from a colon carcinoma and not from the choroidal melanoma. Among 33 patients who completed 5 years of follow-up (31 survivors, 1 death, and 1 lost to follow-up), 2 had melanoma metastasis to the liver (6%). This rate is comparable with the 5-year rate in the COMS medium melanoma study.¹⁹ Metastasis at 10 years in the COMS large tumor trial exceeded 40%.²⁰ Damato et al⁹ reported 1 death due to metastasis at 41 months after surgery. Karkhaneh et al¹⁸ reported 1 death caused by liver metastasis. Kertes et al¹⁷ reported three deaths due to metastasis. It is worthy of note that our local recurrence and metastasis rates are comparable with other studies on endoresection while bearing in mind that our series included patients with larger tumors in diameter and thickness, and a longer follow-up time.

Recently, Konstantinidis et al²¹ reported similar results for metastasis and specific mortality in a large endoresection series, those outcomes were similar to our series. The metastatic mortality was 9% at 5 years and 10 years. These authors suggest that metastasis is

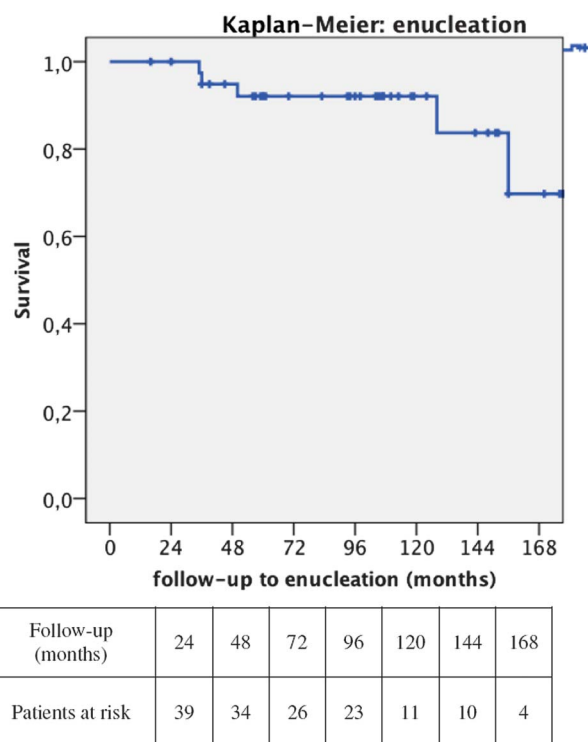


Fig. 7. Kaplan–Meier visual survival curve: enucleation.

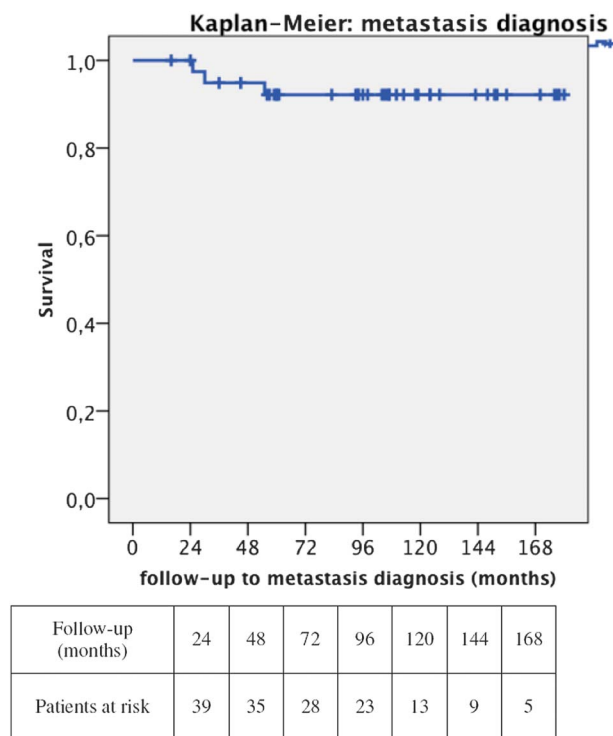


Fig. 8. Kaplan-Meier visual survival curve: metastasis diagnosis.

a biologic process, with almost all Monosomy 3/Class 2 melanomas proving fatal irrespective of any surgical intervention. Their series included 10 patients with Monosomy 3 (14%). In our series, histologic analysis was performed in 21 cases, and Monosomy 3 was detected in 23.8% of patients. This seems to be a small number compared with other authors as Scholes et al²² who detected Monosomy 3 in >50% of their cases. This difference may explain better outcomes of metastasis and survival.

It is worthy of note that the rates of metastasis reported in this study and in other reports should be always interpreted with caution. The reason is that patients whose screening tests reveal negative results for metastasis could still have micrometastatic deposits that could escape detection by laboratory and imaging tests.

In comparison with other techniques, 5-year local recurrence has been reported at 4.2% with Iodine 125 and 5.2% with proton beam radiation.²³ Transscleral local resection had a 42% recurrence rate at 3 years of follow-up.²⁴ The COMS medium tumor study reported a 10.3% 5-year risk of treatment failure using radiotherapy.²⁵ Recently, Papageorgiou et al²⁶ and Russo et al²⁷ reported their results with ¹⁰⁶Ru plaque brachytherapy for posterior choroidal melanomas. In the study by Papageorgiou et al, local recurrence was 7.4% at a mean follow-up of 33.6 months and in the

Russo et al study, local recurrence was 3.7% at a median of 3.5 years. Hadden et al²⁸ found 5 undetected local recurrences in a series of 12 enucleations. In agreement with these authors, we also believe opaque media may contribute to under diagnosis of local recurrence.

In terms of longevity, the initial satisfactory outcome of endoresection studies has always been questioned because most of them include a relatively short follow-up period.⁶ In the present series, the mean follow-up time was 8.5 years, the longest reported in the endoresection literature. The study by Karkhanavich et al¹⁸ had a mean follow-up of 7.4 years.

This study is nonrandomized, which makes comparison with other therapeutic modalities less reliable. However, we can affirm that the local recurrence risk at 5 years in our series is higher than with 125I radiotherapy, proton beam radiation, and ¹⁰⁶Ru plaque brachytherapy, but lower than with transscleral resection and the COMS medium tumor study results.

In an attempt to prevent local recurrence after surgery, we have been performing adjunctive treatment with brachytherapy since January 2003. Brachytherapy seems to inhibit local recurrence by destroying residual malignant cells in the scleral bed and by preventing intrascleral seeding.²⁹ Bechrakis and Foerster³⁰ performed proton beam irradiation in 58 patients before endoresection. The mean follow-up time was 21.7 months, and there were no local recurrences, although 3 patients developed metastasis. Shilling et al³¹ performed stereotactic gamma-knife radiosurgery in 46 cases. Seven patients developed liver metastasis, and 6 patients died over a short follow-up time of just 410 days. In this study, we did not detect a statistically significant difference in terms of 10-year survival between patients who received brachytherapy and those who did not ($P = 0.1$), but it should be kept in mind that the number of patients analyzed for local recurrence was too small (5 patients) for calculating statistical significance. However, there seemed to be a trend toward less local recurrence with combined therapy than with monotherapy (Figure 6), because all 5 patients who had local recurrence were enrolled in the monotherapy group.

This study complements the information our group published in 2001 and 2008. Longer follow-up is perhaps the main new contribution, but other items should be mentioned like the use of concomitant brachytherapy, applied to avoid remnant tumor tissue at the sclera bed. After the longer follow-up, the addition of local radiotherapy seemed to be useful for decreasing the rate of local recurrences.

The primary goal of endoresection is saving the patient's life, the secondary goal is globe salvage, and

the third goal is preservation of visual acuity. Our final visual acuity results are better than those reported in other endoresection series.^{9,17} Visual acuity was $\geq 20/200$ in 48.8% of our patients at the end of follow-up. In another study, using more conservative treatment options for large melanomas, only 5.6% of patients had visual acuity over 20/200 in the brachytherapy group. This rate improved to 61.1% in patients treated with transscleral resection.¹³

Tumor location is also an important factor in determining functional outcome. In this study, all tumors were posterior to the equator, and in 39% of cases, the tumor was temporal to the optic disk. In cases of endoresection for temporally located tumors, the macula and visual function are more likely to be affected by the surgery and its complications than by other techniques used for more anteriorly located tumors such as transscleral resection.

The most serious complication in this study was retinal detachment after silicone oil removal (28.9%). The incidence of this complication in other endoresection studies ranged from 9.4% to 32.6%.^{9,23} The rates of other commonly described complications, such as cataracts, ocular hypertension, and epiretinal proliferation were similar to our results.^{9,11,17,18} Additional procedures are often needed after endoresection, such as silicone oil removal, or cataract surgery. Patients should be informed of this possibility before giving their consent for endoresection.

Although we were not able to carry out genetic study in all cases of resected melanomas in our series, an important advantage of the endoresection technique is that it provides tumor tissue for chromosomal analysis. Increasing evidence from genetic studies suggests that metastasis is a biologic process, with almost all Monosomy 3/Class 2 melanomas proving fatal irrespective of any surgical intervention.³²

Conclusion

Endoresection remains a controversial technique, but it may evade enucleation and preserve vision in large uveal melanomas. Other forms of conservative treatment are likely to cause severe ocular complications, with enucleation being the only alternative. In the present series, long-term follow-up did not show a higher risk of metastasis or local recurrence, and survival rates were similar to other techniques, although comparison is difficult because of the unusual presentation of this type of melanoma. Further studies and longer follow-up are needed to establish the safety of this procedure and to consider endoresection a standard of care and not merely an investigative technique.

Key words: endoresection, choroidal melanoma, internal resection of uveal melanoma, local resection of choroidal melanoma, ab interno retinchoroidectomy.

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